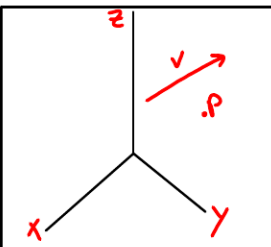


Ameet Rao's Calculus III Notes

Equations of Lines and Planes, Angles Between Planes, and Point-Line-Plane Distances



$v = \langle a, b, c \rangle$
 $P(x, y, z)$
 "slope" and point

Symmetric
 $\frac{x-x_1}{a} = \frac{y-y_1}{b} = \frac{z-z_1}{c}$

Parametric
 $x = x_1 + at$
 $y = y_1 + bt$
 $z = z_1 + ct$

A line needs a point and a direction $v = \langle a, b, c \rangle$. Parametric form moves t units along v ; solving each equation for t gives the symmetric form.

$P(1, -2, 4)$ $v = \langle 2, 4, -4 \rangle$

Symmetric
 $\frac{x-1}{2} = \frac{y+2}{4} = \frac{z-4}{-4}$

Parametric
 $x = 1 + 2t$
 $y = -2 + 4t$
 $z = 4 - 4t$

Read the point and direction straight off: $P(1, -2, 4)$ and $v = \langle 2, 4, -4 \rangle$ give $x = 1 + 2t$, $y = -2 + 4t$, $z = 4 - 4t$.

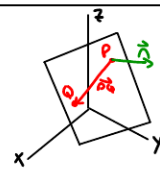
$P(-2, 0, 0)$ $Q(1, 3, 5)$

$PQ = \langle 3, 3, 5 \rangle$

$\frac{x+2}{3} = \frac{y}{3} = \frac{z}{5}$

$x = -2 + 3t$
 $y = 3 + 3t$
 $z = 5 + 5t$

Given two points, PQ is the direction vector: $PQ = \langle 3, 3, 5 \rangle$. Then use either point in the symmetric or parametric form.



P, Q points on Plane
 $\vec{n}$ (normal vector) $\perp \vec{PQ}$
 $n = \langle a, b, c \rangle$
 $P(x_1, y_1, z_1)$ $Q(x_2, y_2, z_2)$

$n \cdot PQ = 0$

$\langle a, b, c \rangle \cdot \langle (x_2 - x_1), (y_2 - y_1), (z_2 - z_1) \rangle$

Standard Form
 $a(x_2 - x_1) + b(y_2 - y_1) + c(z_2 - z_1) = 0$

$ax_2 + by_2 + cz_2 = 0$

P is known
 Q is NOT known
 General Form
 $ax + by + cz + d = 0$

$d = -ax_1 - by_1 - cz_1$

A plane needs a point and a normal vector n . Any PQ lying in the plane is perpendicular to n , so $n \cdot PQ = 0$. Expand for standard form; collect constants for $d = -(ax_1 + by_1 + cz_1)$.

Find eq. of plane containing
 $P(2, 1, 1)$ $Q(0, 4, 1)$ $R(-2, 1, 4)$

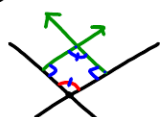
$\vec{n} = \langle 9, 6, 12 \rangle$

$9(x-2) + 6(y-1) + 12(z-1) = 0$

$9x + 6y + 12z - 36 = 0$

Three points make a plane. Build PQ and PR , take the cross product for $n = \langle 9, 6, 12 \rangle$, then use n and P : $9x + 6y + 12z = 36$.

Angle between two planes



$\cos \theta = \frac{n_1 \cdot n_2}{\|n_1\| \|n_2\|}$

If $n_1 \cdot n_2 = 0$ Planes $\perp$

If $n_1 = c n_2$ Planes Parallel

The angle between planes is the angle between their normals: $\cos \theta = n_1 \cdot n_2 / (\|n_1\| \|n_2\|)$. $n_1 \cdot n_2 = 0$ means perpendicular; $n_1 = c n_2$ means parallel.

Ameet Rao's Calculus III Notes

Equations of Lines and Planes, Angles Between Planes, and Point-Line-Plane Distances

Distance between Point and Plane



$$D = \frac{|PQ \cdot n|}{\|n\|}$$

Point to plane: project PQ onto the normal n , with P any point on the plane and Q the outside point. $D = |PQ \cdot n| / \|n\|$.

$$Q(1, 5, -4) \quad 3x - y + 2z = 6$$

$$PQ = \langle 1, 5, 4 \rangle$$

$$n = \langle 3, -1, 2 \rangle$$

$$PQ \cdot n = 3 + 5 + 8 = 16$$

$$P(2, 0, 0)$$

$$\|n\| = \sqrt{3^2 + (-1)^2 + 2^2} = \sqrt{14}$$

$$D = \frac{16}{\sqrt{14}}$$

Plane $3x - y + 2z = 6$ gives $n = \langle 3, -1, 2 \rangle$. With $P(2, 0, 0)$ on the plane, $|PQ \cdot n| = 16$ and $\|n\| = \sqrt{14}$, so $D = 16/\sqrt{14} \approx 4.28$.

Distance between Point and line

P = Point on line

Q = Point NOT on line

u = direction of line

$$D = \frac{\|PQ \times u\|}{\|u\|}$$

Point to line: $\|PQ \times u\|$ is a parallelogram's area and $\|u\|$ is its base, so dividing gives the height, the shortest distance to the line.

Distance between two parallel planes

$$D = \frac{|ax_0 + by_0 + cz_0 + d|}{\sqrt{a^2 + b^2 + c^2}}$$

Parallel planes: pick any point (x_0, y_0, z_0) on one, plug it into the other's $ax + by + cz + d$, and divide by $\sqrt{a^2 + b^2 + c^2}$.